

The Place or the Pieces?
Evaluating the Impact of Bike Infrastructure and External Factors in Cyclist Crashes

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Abstract

This study examines the impact of bicycle infrastructure and external factors on cyclist crash severity in North Carolina, using spatial analysis and logistic regression models on a dataset of 15,573 police-reported crashes. Results indicate that sidewalks and crosswalks significantly reduce severe injury risk compared to travel lanes, while multi-use paths and bike lanes show higher crash rates per mile. Rural areas emerge as hotspots for severe crashes, with road classification, lighting, and weather conditions, particularly fog, playing critical roles in injury outcomes. Despite data limitations, the findings highlight the need for targeted investments in safer cycling infrastructure, improved data accessibility to better inform planning decisions, and to address disparities between urban and rural settings.

A. Introduction

The United States of America is a car-centric society. This isn't new information, as the prevalence of roads and the 228,915,520 licensed drivers across this nation makes it abundantly clear.¹ However, the side effect of this type of society that prioritizes cars is the endangerment of pedestrians, particularly cyclists, as they have to deal with subpar infrastructure situations that endanger their lives. This can best be seen in Harrisburg, North Carolina, where there is a single designated bike lane that fails to provide meaningful connectivity or ensure safety over major roads like NC Highway 49 to the town's nineteen thousand residents.² These shortcomings reflect a statewide trend: Only 0.2% of North Carolinians commute by bike, just one-third of the national 0.6% rate.³ This correlates to the fatality rate, where North Carolina is under the national average of 0.33 cyclist fatalities per 100,000 people, with its rate of 0.21.⁴ However, when accounting for exposure, North Carolina ranks among the top 20 states in cyclist fatalities per 1,000 cyclists, with a rate of 1.86 deaths per 1,000 riders.⁵

This disparity in bicycle usage is reflected in the state's infrastructure, with only 3,463 miles designated for bicyclists out of the 21,632 miles in the Pedestrian and Bicycle Infrastructure Network (PBIN).⁶ This underinvestment coincides with higher-than-average fatality rates per cyclist, fueling ongoing debates about infrastructure safety. This issue was highlighted by Alan Wachtel and Diana Lewiston, who found that bicyclists are 1.8 times more likely to be injured on sidewalks compared to roadways.⁷ In the case of North Carolina, poor bike infrastructure leads to more injuries and discourages cycling, since safety concerns are a

¹U.S. Department of Transportation, Federal Highway Administration, Highway Statistics 2019: Licensed Drivers, by Sex and Percentage in Each Age Group and Relation to Population, Table DL-1C, last modified February 2021, <https://www.fhwa.dot.gov/policyinformation/statistics/2019/dl1c.cfm>.

²U.S. Census Bureau, QuickFacts: Harrisburg town, North Carolina, last modified 2023, <https://www.census.gov/quickfacts/fact/table/harrisburgtownnorthcarolina/PST045223>.

³"Existing Conditions : Walk Bike NC," n.d., https://www.ncdot.gov/bikeped/walkbikenc/mobility_post/existing-conditions/default.aspx.

⁴National Center for Statistics and Analysis, Bicyclists and Other Cyclists: 2022 Data, Traffic Safety Facts, Report No. DOT HS 813 591 (Washington, DC: National Highway Traffic Safety Administration, July 2024), 10.

⁵Bader Scott Injury Lawyers, Bicycle Accidents Study 2025: The Most and Least Dangerous States, accessed April 19, 2025, <https://baderscott.com/research/bicycle-accidents-study-2025>

⁶"Pedestrian and Bicycle Infrastructure Network (PBIN) | Connect NCDOT," n.d., <https://connect.ncdot.gov/projects/BikePed/pages/pbin.aspx>.

⁷Alan Wachtel and Diana Lewiston, "Risk Factors for Bicycle-Motor Vehicle Collisions at Intersections," ITE Journal 64, no. 9 (September 1994): <https://www.bicyclinglife.com/Library/Accident-Study.pdf>. Pp 8.

major reason people avoid biking.⁸ However, a study within Toronto and Ottawa suggests that the issue extends beyond infrastructure alone. Rather, cyclists who use pedestrian-oriented infrastructure, such as sidewalks, tend to be less experienced than those who ride on roads.⁹ This was proven by the fact that primarily sidewalk cyclists had more crashes on the road than non-sidewalk cyclists.¹⁰ This corresponds with Hee et al., who argue that increased bicycle volume in dedicated lanes could inflate crash rates, with the most common cause of injuries being inexperienced cyclists falling on their own.^{11 12}

Teschke et al. (2012) also found that sidewalks improve safety in the absence of bike infrastructure, though other infrastructure types offer greater safety.¹³ They accounted for variables like traffic and weather to control for divergence, rather than to measure their contribution to crashes.¹⁴ However, the differing results between Ottawa and Toronto prompted both studies to call for further research with broader regional datasets with a greater number of observations.^{15 16}

North Carolina's Department of Transportation (NCDOT) crash dataset, which documents 51,415 non-motorist incidents, provides a unique opportunity to resolve ongoing debates surrounding bicycle safety and infrastructure design.¹⁷ Unlike prior studies that often relied on fewer than a thousand observations limited to urban areas like Toronto, this dataset offers a statewide scope that captures variation in urban, suburban, and rural settings. This study fills existing gaps by employing binomial and multinomial logistic models to assess how

⁸Teschke et al., "Route Infrastructure and the Risk of Injuries to Bicyclists: A Case-Crossover Study."

⁹Lisa Aultman-Hall and Michael F. Adams, "Sidewalk Bicycling Safety Issues," *Transportation Research Record Journal of the Transportation Research Board* 1636, no. 1 (January 1, 1998): 71–76, <https://doi.org/10.3141/1636-11>. Pp 75.

¹⁰Aultman-Hall and Adams, "Sidewalk Bicycling Safety Issues." Pp 75.

¹¹Jung Hee Wee et al., "A Comparative Study of Bike Lane Injuries," *Journal of Trauma and Acute Care Surgery* 72, no. 2 (February 1, 2012): 448–53, <https://doi.org/10.1097/ta.0b013e31823c5868>. Pp 450.

¹²Wee et al., "A Comparative Study of Bike Lane Injuries." Pp 452.

¹³Teschke et al., "Route Infrastructure and the Risk of Injuries to Bicyclists: A Case-Crossover Study." Pp 5.

¹⁴Teschke et al., "Route Infrastructure and the Risk of Injuries to Bicyclists: A Case-Crossover Study." Pp 6.

¹⁵Aultman-Hall and Adams, "Sidewalk Bicycling Safety Issues." Pp 75.

¹⁶Teschke et al., "Route Infrastructure and the Risk of Injuries to Bicyclists: A Case-Crossover Study." Pp 6.

¹⁷North Carolina Department of Transportation, Non-Motorist Crashes in NC from 2007 to 2023, GIS dataset, ArcGIS, October 22, 2024, <https://ncdot.maps.arcgis.com/home/item.html?id=d00e97d23146472cb5da78fc3dbc0274>

infrastructure type (bike lanes, sidewalks, travel lanes, etc) and environmental factors (light, weather, road surface) interact to influence severity.

By integrating prior findings with a regional dataset, this project aims to move beyond the limitations of case-specific studies and provide actionable insights for state and local policymakers who deal with 77.5% of the total US public infrastructure spending.¹⁸ The goal is to identify which types of infrastructure truly lower the severity under the various conditions faced by cyclists, not just under ideal or isolated circumstances. These findings will support the development of more effective infrastructure systems and promote cycling as a viable and safe mode of transportation across North Carolina and similar contexts.

B. Data Sources and Limitations

This study primarily relies upon the NCDOT's Non-Motorist Crashes dataset and the NCDOT's PBIN shapefile. The former dataset includes police-reported bicycle crashes from January 2007 to December 2023, which includes 15,573 bicycle-related crashes out of the 51,415 total non-motorist crashes. Using this dataset will lead to inaccuracies, as many bicycle crashes are not reported to police databases, even if there is a major injury that occurs.¹⁹ Additionally, NC reporting laws don't allow the dataset to contain any crashes that do not involve a motor vehicle in motion, nor do they allow for crashes that occurred within private property.²⁰ Additionally, since this isn't survey data like the Toronto and Vancouver report, minor crashes such as noninjury incidents or cases where the trip was not continued are likely underreported, as they may not warrant police intervention.²¹ The NCDOT PBIN dataset inventories bicycle and pedestrian infrastructure, but is outdated, last updated in July 2024. As a crowdsourced resource

¹⁸Joseph W. Kane, Adie Tomer, and Ben Swedberg, "Four Recent Trends in US Public Infrastructure Spending," Brookings, March 27, 2025, <https://www.brookings.edu/articles/four-recent-trends-in-us-public-infrastructure-spending/>.

¹⁹Aultman-Hall and Adams, "Sidewalk Bicycling Safety Issues." Pp 71.

²⁰North Carolina Department of Transportation, Non-Motorist Crashes in NC from 2007 to 2023 (GIS dataset; ArcGIS, October 22, 2024), <https://ncdot.maps.arcgis.com/home/item.html?id=d00e97d23146472cb5da78fc3dbc0274>.

²¹Peter A. Crompton et al., "Severity of Urban Cycling Injuries and the Relationship with Personal, Trip, Route and Crash Characteristics: Analyses Using Four Severity Metrics," *BMJ Open* 5, no. 1 (2015): e006654, <https://doi.org/10.1136/bmjopen-2014-006654>.

relying on public input, it is prone to inconsistencies, omissions, and inaccuracies. This was evident in the underrepresentation of multi-use paths and errors like a mapped sidewalk appearing in the middle of the Atlantic Ocean.

A final note on data availability is the significant lack of publicly accessible exposure data. Ideally, this would come from sources like the North Carolina Non-Motorized Volume Data Program, which uses sensors and cameras to count bicycles and pedestrians at key intersections and facilities across the state.²² If made public, such data would strengthen this study's findings. Additionally, accurate county-level data on bicycle infrastructure and biking population estimates would also support future research. While the 2023 American Community Survey included bike commuting rates at the county level, several counties with reported crashes still showed 0.0% biking to work.²³ This highlights a major challenge: the lack of current, accurate, and accessible data. With more time, a researcher could manually collect this data using methods like Alan Wachtel's intersection-based observational counts, but such an approach was not feasible for this study.²⁴

C. Methodology

a. ArcGIS Spatial Analysis

This analysis begins by standardizing the coordinate reference system to ensure consistency across all datasets. The chosen CRS was NAD 1983 State Plane NC FIPS 3200 (US Feet), matching the CRS of the count data and allowing for measurements in the imperial system. This will be important for direct measurements of the PBIN data.

To focus on bicycle-related crashes, the non-motorist count dataset was filtered using the Select by Attribute tool, isolating entries where "NM_Type" was "Bicyclist." This subset,

²² Institute for Transportation Research and Education, North Carolina Non-Motorized Volume Data Program (NC NMVDP), North Carolina State University, accessed April 19, 2025, <https://itre.ncsu.edu/focus/bike-ped/nc-nmvdpl/>.

²³ U.S. Census Bureau, U.S. Department of Commerce, "Commuting Characteristics by Sex," 2023, American Community Survey, ACS 1-Year Estimates Subject Tables, Table S0801, accessed March 17, 2025, <https://data.census.gov/table/ACSST1Y2023.S0801?q=Commuting+Data+By+Sex>.

²⁴ Alan Wachtel and Diana Lewiston, "Risk Factors for Bicycle-Motor Vehicle Collisions at Intersections," ITE Journal 64, no. 9 (September 1994). Pp 3.

exported into a new feature, contained 15,573 crash records. To simplify crash locations and link them more clearly to infrastructure types, a new field, “Simplified_Pos2,” was created using the Field Calculator. This field reclassified the “NM_Pos” values into broader categories: “Bike Lane/Paved Shoulder,” “Sidewalk/Crosswalk,” “Travel Lane,” “Multi-Use Path,” “Driveway/Other,” and “Other.” Grouping similar descriptors made it easier to identify meaningful patterns by increasing the number of observations per category.²⁵

To assess crash severity spatially, two weighted scoring systems were applied, drawing from the Florida Department of Transportation’s study of bicycle injury hotspots. While it does not replicate their full network screening methodology, it adopts a comparable injury severity weighting system.^{26 27} The main difference between the two hot-spot analyses is that the first weighted system follows the FDOT study, with a focus on high-severity clusters.²⁸ To do this, the study assigned numerical values based on injury severity: 3 (Killed), 2 (Suspected Serious Injury), and 1 (Suspected Minor Injury). The second weighting system expanded the weight to 4, with 1 (Possible Injury) and 0 (No Injury) being added. These weighted scores facilitated hotspot analysis to identify regions with disproportionately high-severity crashes. Additionally, the study created a Kernel Density map showcasing all of the crashes to highlight high-incidence areas.

The PBIN shapefile was imported to examine bicycle infrastructure distribution. At this stage, previously noted data quality issues became more apparent, limiting the depth of analysis. Despite these challenges, two Kernel Density maps were produced to visualize infrastructure exposure across the state. Additionally, total lengths (in miles) of multi-use paths, sidewalks, and bike lanes were calculated and exported to R, along with the complete bicycle crash dataset.

b. Logistic Regression

²⁵Priyanka Alluri et al., *Statewide Analysis of Bicycle Crashes* (Tallahassee, FL: Florida Department of Transportation, 2017), Pp. vii, 75-76.

²⁶Priyanka Alluri et al., *Statewide Analysis of Bicycle Crashes* (Tallahassee, FL: Florida Department of Transportation, 2017), <https://rosap.nhtl.bts.gov/view/dot/32431>. Pp.vi, vii.

²⁷Priyanka Alluri et al., *Statewide Analysis of Bicycle Crashes* (Tallahassee, FL: Florida Department of Transportation, 2017), Pp. 68–69.

²⁸Priyanka Alluri et al., *Statewide Analysis of Bicycle Crashes* (Tallahassee, FL: Florida Department of Transportation, 2017), Pp. ix, 108-117.

Once exported from ArcGIS, the study now transitions into a two-pronged statistical modeling approach within R to primarily assess how roadway design and external factors influence crash severity outcomes for bicyclists. Two types of models were used to explore these relationships: binary logistic regression and multinomial logistic regression.

Before that, the crash dataset was imported and organized to simplify and standardize similar variables. A new variable for severity classification was generated to consolidate injury levels into three tiers: “High” (including fatalities and suspected serious injuries), “Moderate” (including suspected minor and possible injuries), and “Low” (no injury). Additionally, a binary severity variable was created to isolate high-severity crashes, facilitating logistic regression. A binary version of this variable was also created to distinguish severe crashes from non-severe ones, which was used in the logistic regression model. To simplify the analysis and expand the number of observations in underrepresented crash scenarios, environmental and roadway characteristics such as lighting, weather, surface type, and road condition were grouped into broader categories based on common traits.

This leads to the binomial logistic regression model, which was constructed to predict the probability of “High” injury crashes based on infrastructure type while controlling for environmental and roadway factors. Reference categories used in the model were those with the highest number of observations, such as “Travel Lane,” “Dry,” “Clear,” “Smooth Asphalt,” and “Daylight.” Covariates included lighting, weather, road classification, surface condition, crash location type, and number of travel lanes.

This was loosely based on the work by Crompton et al., who analyzed cycling injuries in Toronto and Vancouver using four different severity metrics.²⁹ Their analysis linked factors such

²⁹Peter A. Crompton et al., “Severity of Urban Cycling Injuries and the Relationship with Personal, Trip, Route and Crash Characteristics: Analyses Using Four Severity Metrics,” *BMJ Open* 5, no. 1 (2015).

as route type and motor vehicle involvement to injury severity.³⁰ While inspired by their framework, this study differs in several key ways: it utilizes police-reported crash data from North Carolina rather than self-reported hospital records, relies on official crash report classifications instead of clinical triage scales, and incorporates additional variables such as number of lanes, road surface condition, surface type, and, most notably, weather.³¹

To capture a fuller picture of injury outcomes, a multinomial logistic regression model was also created. Unlike the binary model, this model estimated the probabilities for each severity level: low, moderate, and high, allowing for more nuanced comparisons across infrastructure types. The same set of predictors was used, and the results were visualized to better illustrate how injury severity is distributed across different crash scenarios.

The results of these models were visualized using the `ggpredict` function in R, which showed how the risk of injury varies by infrastructure type. These visuals were further broken down by conditions like lighting, weather, road surface, and traffic characteristics to explore how the impact of infrastructure shifts in different environments.

To make the models more relatable, the analysis also looked at basic stats like how crashes broke down by infrastructure type, how severe they were, and how often they happened per mile based on PBIN data. Heatmaps further visualized how injury severity varies by infrastructure and road classification, showcasing injury-prone areas.

D. Results

a. Spatial Patterns of Bicycle Crashes and Pedestrian Infrastructure in ArcGIS

³⁰Peter A. Crompton et al., “Severity of Urban Cycling Injuries and the Relationship with Personal, Trip, Route and Crash Characteristics: Analyses Using Four Severity Metrics,” *BMJ Open* 5, no. 1 (2015). Pp.4.

³¹Peter A. Crompton et al., “Severity of Urban Cycling Injuries and the Relationship with Personal, Trip, Route and Crash Characteristics: Analyses Using Four Severity Metrics,” *BMJ Open* 5, no. 1 (2015). Pp.6.

Figure One - Bike Severity Hot Spot Analysis (Severe)

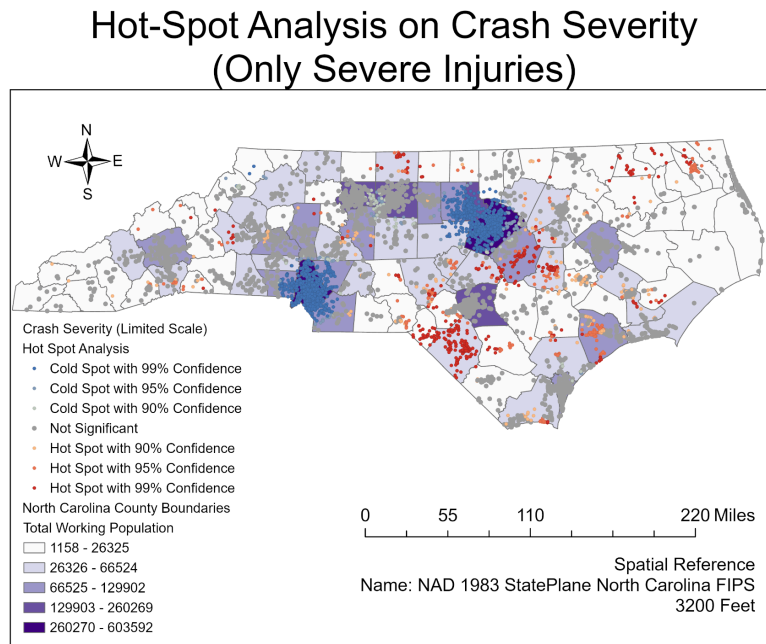


Figure Two- Bike Severity Hot Spot Analysis (All Injuries)

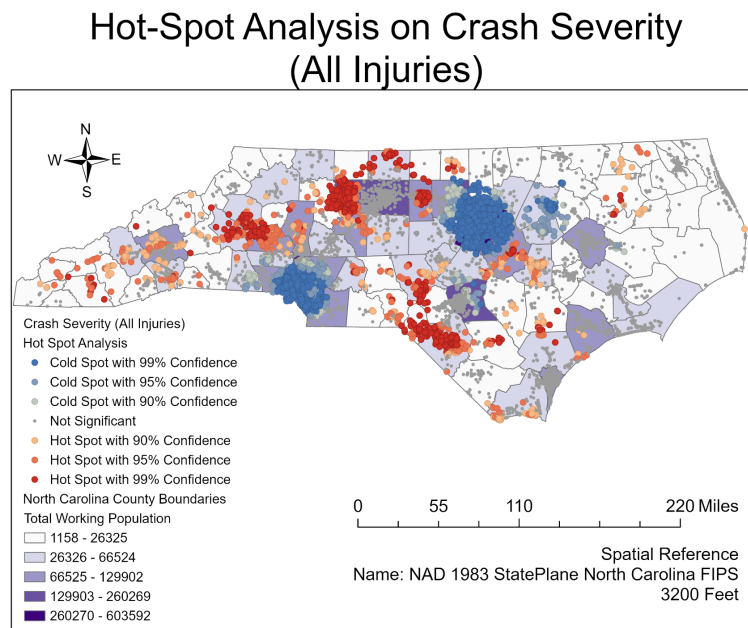
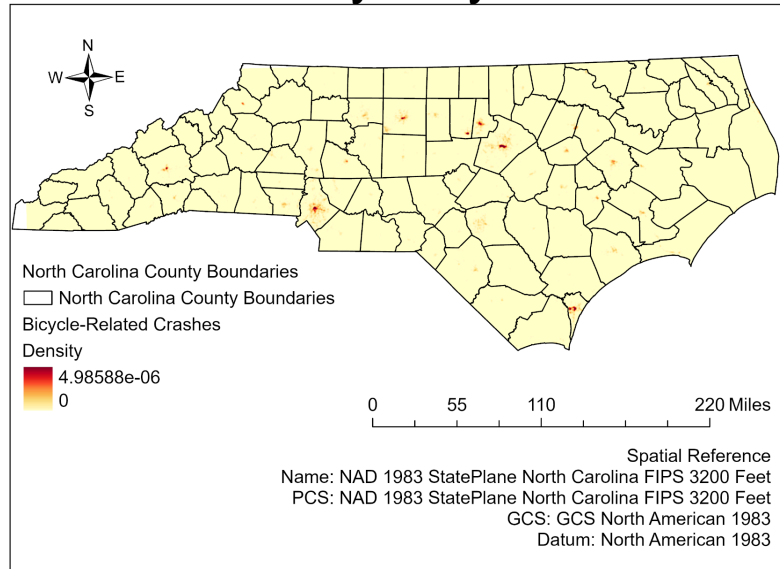


Figure Three- Bike Crash Density Map

Kernel Density Bicycle Crashes

*Figure Four- Bike Facilities Density Map*

Kernel Density Bicycle Facilities

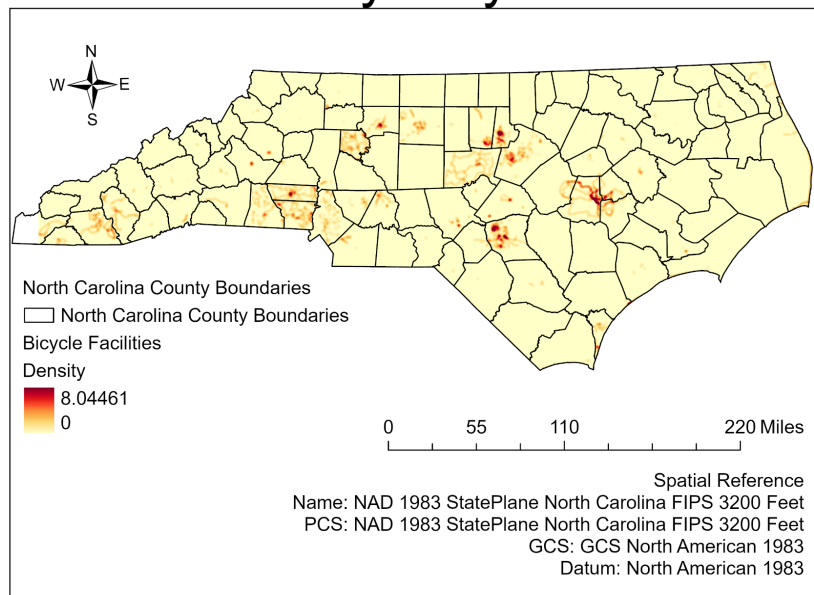
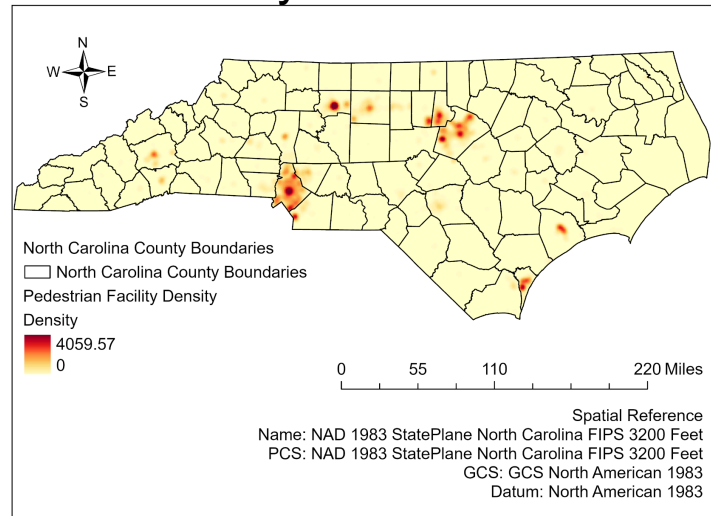


Figure Five - Pedestrian Facilities Density Map

Kernel Density Pedestrian Facilities



The spatial analysis revealed distinct geographic patterns in bicycle crash severity and distribution across North Carolina. The hot spot analysis for both Figures 1 and 2 identified consistent cold spots within the states' major urban areas, such as Raleigh and Charlotte. In contrast, notable hot spots for all injury crashes were found in less populated counties such as Bladen, Robeson, Scotland, and Richmond. When isolating severe injury crashes, hot spots were concentrated in counties like Robeson, Johnston, Hertford, Gates, and Camden. Overall, there were fewer confident hot spots for severe injuries compared to all injuries.

Figures 3, 4, and 5 present the second half of the spatial analysis, highlighting crash and infrastructure density using Kernel Density Estimation. Figure 3 displays the primary areas where bicycle crashes occurred, which include urban centers such as Raleigh, Durham, Greensboro, Chapel Hill, Charlotte, and Wilmington. Figure 4 illustrates the concentration of bicycle facilities, found in places like Goldsboro, Kinston, Fayetteville, Lincolnton, and Durham. Finally, Figure 5 shows the distribution of pedestrian facilities (primarily sidewalks), which

included urban centers such as Jacksonville, Wilmington, Charlotte, Raleigh, Durham, Greensboro, and Asheville.

b. Basic R Observations

Figure Six - Severity Distribution (Non-Model)

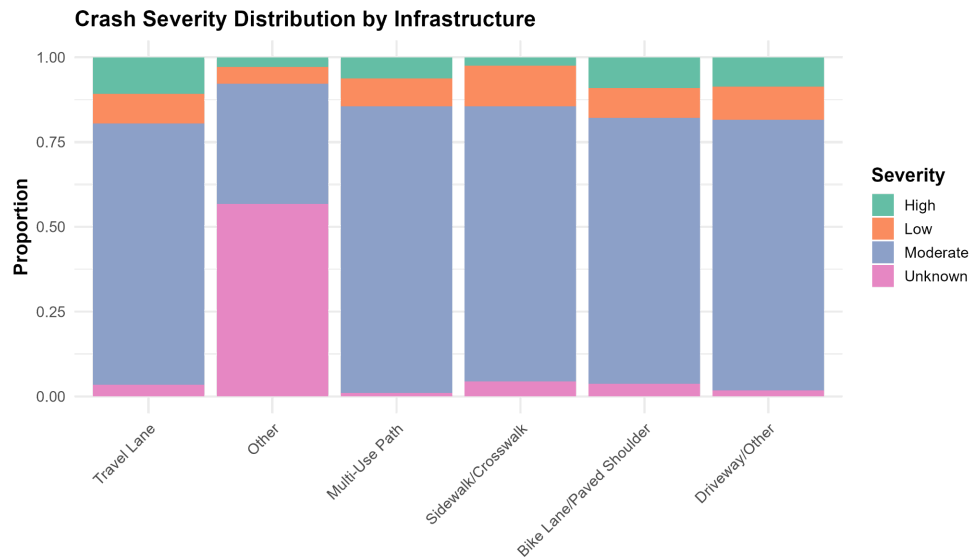


Figure Seven - Crash Frequency By Infrastructure

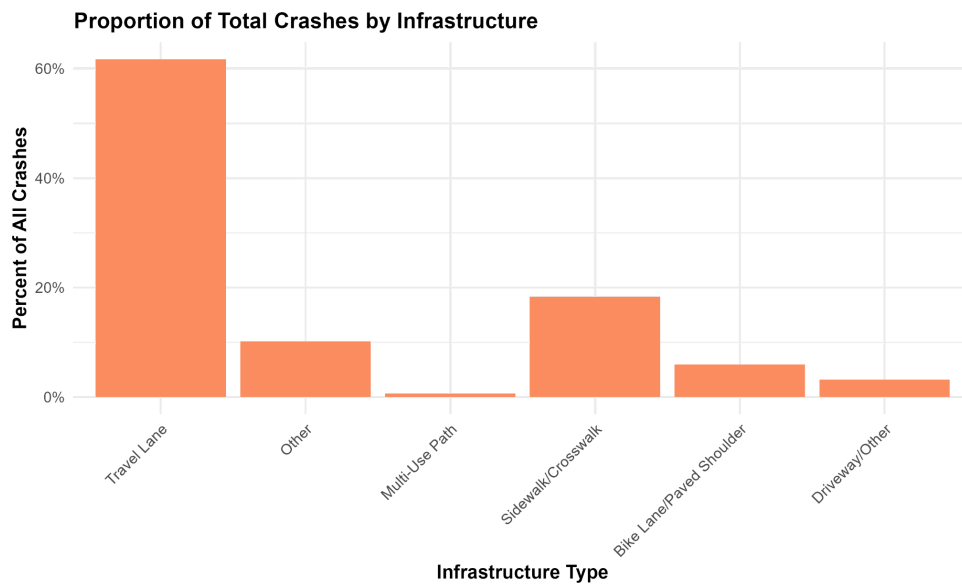
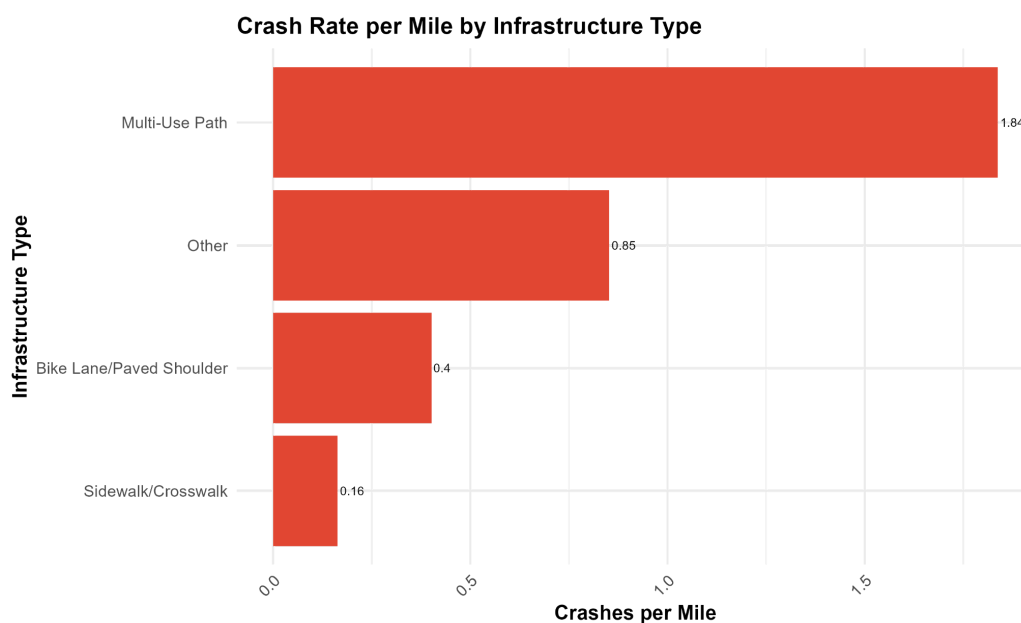
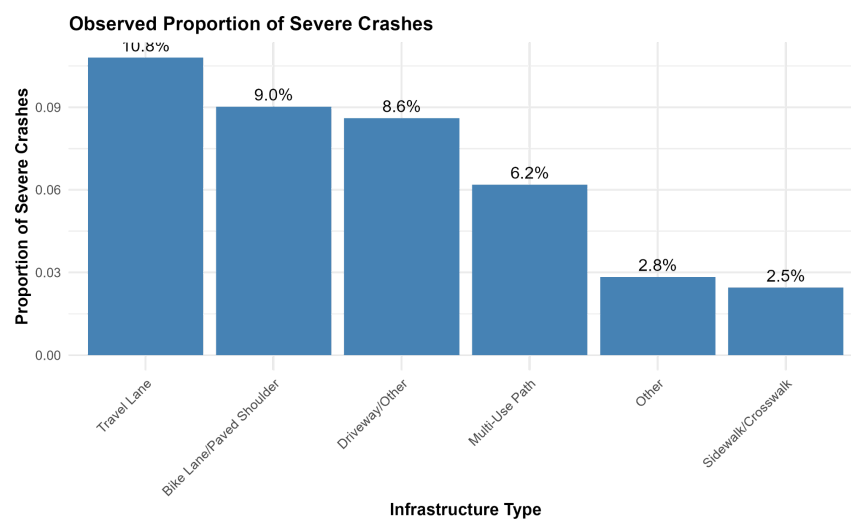


Figure Eight - Crash Rate Table

Simplified_Pos2	Crashes	Miles	CrashRatePerMile
Other	1,588	1,863.5874	0.8521199
Multi-Use Path	97	52.7986	1.8371699
Sidewalk/Crosswalk	2,852	17,427.7951	0.1636466
Bike Lane/Paved Shoulder	920	2,288.3328	0.4020394

Figure Nine - Crash Rate Per Mile*Figure Ten - Observed Proportion of Severe Crashes by Facility*

This part of the analysis begins with displaying the distribution of both crash severity and total amount of crashes by infrastructure type (Figures 6 & 7). These culminated in ‘Travel Lane’ having the highest proportionate amount of highly severe crashes (10.8%) as well as the most amount of crashes total, with 61.7% of the crashes being credited to it. Multi-use path, Bike lane/Paved Shoulder, and Driveway/Other had similar distributions of severity (High - 9%, Low 8.8%, Medium - 78.5%). While Sidewalk/Crosswalk had the highest proportion of low-severity crashes (12.0%). Sidewalk/Crosswalk made the second most amount of crashes (18.3%), then Bike Lane/Paved Shoulder (5.9%), then Driveway/Other (3.2%), then Multi-use Pathways (0.6%). This changes when comparing the crash rate by mile, with Figures 8 & 9, which had Multi-use path in the lead (crashes 1.84 per mile), then Bike Lane/Paved Shoulder (crashes 0.4 per mile), and then Sidewalk/Paved Shoulder (crashes 0.16 per mile). Figure 10 displays the amount of severe crashes by facility: Travel Lane (10.8%), Bike Lane (9.0%), Driveway/Other (8.6%), Multi-Use Path (6.2%), Sidewalk/Crosswalk (2.5%).

c. Multinomial Regression Model

Figure Eleven- Multinomial Logistic Regression Summary

Response	Predictor	Coefficient	Std_Error	z_value	p_value	Significance
Moderate	(Intercept)	2.2965702699	0.0814580766225815739	28.193278864	0.00000000000000000000	***
Moderate	Simplified_Pos2Other	-0.1644430242	0.1314302237160592524	-1.251181193	0.210868378036231929329	
Moderate	Simplified_Pos2Multi-Use Path	0.1284341900	0.3742353794242252452	0.343190936	0.731454823387500896459	
Moderate	Simplified_Pos2Sidewalk/Crosswalk	-0.2678572586	0.0755276091845559633	-3.546481366	0.000390412351996438645	***
Moderate	Simplified_Pos2Bike Lane/Paved Shoulder	-0.0009157078	0.1252554354946676940	-0.007310723	0.994166938896944740733	
Moderate	Simplified_Pos2Driveway/Other	-0.0193684395	0.1604429968086576175	-0.120718510	0.903913997604269603414	
Moderate	LightCondDark - Lighted Roadway	0.0899999390	0.0935658774626294265	0.961888473	0.336105629421096452347	
Moderate	LightCondDark - Roadway Not Lighted	0.3115779338	0.1198887708710067307	2.598891719	0.009352526963285257011	**
Moderate	LightCondDusk	0.0306177979	0.1655828023806488469	0.184909287	0.853300186771416058207	
Moderate	LightCondOther	0.2683802535	0.2393198890677273871	1.121428956	0.262105316232427165701	
Moderate	WeatherCloudy	0.0604045164	0.0998585793945122591	0.604900618	0.545245032382988270570	
Moderate	WeatherFog, Smog, Smoke	-0.9800511029	0.5693489947400345308	-1.721353883	0.085186627297197503594	
Moderate	WeatherRain	0.0357558368	0.2219439604722449333	0.161102995	0.872012281107805709013	
Moderate	WeatherOther	-0.7260085837	0.4985796807480162296	-1.456153573	0.145350170458249827732	
Moderate	RdClassNc Route	-0.0272452377	0.0978868697714809777	-0.278333936	0.780756027802806285010	
Moderate	RdClassSecondary Route	0.0078728958	0.0712080201704227700	0.110561925	0.911963741693938567678	
Moderate	RdClassUs Route	0.1153282183	0.0993480950777938521	1.160849820	0.245702978657477366653	
Moderate	RdClassOther	0.0841359526	0.1583444825108033527	0.531347548	0.595177961542250555382	
Moderate	RdSurfaceCoarse Asphalt	-0.1475406508	0.0619772465375046377	-2.380561562	0.017286272367957433715	*
Moderate	RdSurfaceNot Coded	0.0000000000	0.000000000004143058	0.000000000	1.000000000000000000000	
Moderate	RdSurfaceOther	0.0009960320	0.1772617163203926427	0.005618991	0.995516717594609357178	
Moderate	RdConditioUnknown	0.3584185420	0.6604066277487249392	0.542724023	0.587319831220794341675	
Moderate	RdConditioWet	0.0217331514	0.1740804732275953171	0.124845429	0.900645921415282213474	
Moderate	RdConditioOther	-0.5041234020	0.4068584827321358732	-1.239063270	0.215322068001578825758	
Moderate	CrashLocIntersection-Related	-0.3936599672	0.1022066414923257782	-3.851608481	0.000117344526854301634	***
Moderate	CrashLocNon-Intersection	-0.1000637956	0.0658733868676753820	-1.519032198	0.128754391660413247678	
Moderate	CrashLocOther	-0.6062331379	0.6488234139867249395	-0.934357677	0.350119421673509734205	
Moderate	NumLanesLow (1-2)	-0.0599417476	0.0644673289640211172	-0.929800389	0.352474444796825547854	
High	(Intercept)	-0.7931043537	0.1232135396471720629	-6.436827932	0.000000000121996190927	***
High	Simplified_Pos2Other	-0.6976114521	0.2007916725814160031	-3.474304702	0.000512179057486017442	***
High	Simplified_Pos2Multi-Use Path	-0.2678627210	0.5455106192681526966	-0.491031176	0.623404397478683147682	
High	Simplified_Pos2Sidewalk/Crosswalk	-1.4105322455	0.1502398581587988668	-9.388535525	0.000000000000000000000	***
High	Simplified_Pos2Bike Lane/Paved Shoulder	-0.3022213942	0.1680782502012060087	-1.798099361	0.072161263544481046850	
High	Simplified_Pos2Driveway/Other	-0.2118326982	0.2204879122081372933	-0.960745177	0.336680309970089242100	
High	LightCondDark - Lighted Roadway	0.4702235204	0.1278961516513359986	3.676604138	0.000236359391535767571	***
High	LightCondDark - Roadway Not Lighted	1.0700407446	0.1367005248885119206	7.827627184	0.000000000000004884981	***
High	LightCondDusk	0.5408608299	0.2131170582327527874	2.537857994	0.011153323069533760403	*
High	LightCondOther	0.5532844934	0.3048228827967814358	1.815101571	0.069508301867550192910	
High	WeatherCloudy	0.0753424323	0.1380400572749009291	0.545801224	0.585202581653361253444	
High	WeatherFog, Smog, Smoke	0.0464852046	0.6299510781014798777	0.073791769	0.941176076297008723515	
High	WeatherRain	-0.1320267530	0.2972303451585359690	-0.444190020	0.656905207038771310479	
High	WeatherOther	-1.4328082053	0.8813251384243501185	-1.625743035	0.104004331071083200300	
High	RdClassNc Route	0.9192927411	0.1330495873119577621	6.909399418	0.00000000004866995695	***
High	RdClassSecondary Route	0.6809787537	0.1057652235192179796	6.438588517	0.000000000120589538355	***
High	RdClassUs Route	0.9572513618	0.1368254368436847557	6.996150598	0.000000000002631006524	***
High	RdClassOther	-0.8367772879	0.3729202420153174269	-2.243850544	0.024842019100260515074	*
High	RdSurfaceCoarse Asphalt	0.0315901879	0.0840114825488341294	0.376022265	0.706900343829647948013	
High	RdSurfaceNot Coded	0.0000000000				
High	RdSurfaceOther	-0.1396553401	0.2823325361498651986	-0.494648410	0.620848323917583932996	
High	RdConditioUnknown	0.2696232629	0.9805361283169716780	0.274975348	0.783335178132527731876	
High	RdConditioWet	0.0166614086	0.2308314687811389432	0.072179970	0.942458684965266524358	
High	RdConditioOther	-0.2553578742	0.5734996982946870547	-0.445262439	0.656130108987882154636	
High	CrashLocIntersection-Related	-0.2279159736	0.1543563689291657381	-1.476556978	0.13979443127274755487	
High	CrashLocNon-Intersection	0.3274588317	0.0921698108185470683	3.552777518	0.000381186706374769813	***
High	CrashLocOther	-5.6974750849	12.7211339052623078061	-0.447874783	0.654243568602322467243	
High	NumLanesLow (1-2)	0.0521059592	0.0909123661451350096	0.573144902	0.566546586556905706900	

Figure Twelve - Multinomial Logistic Regression (ggpredict)

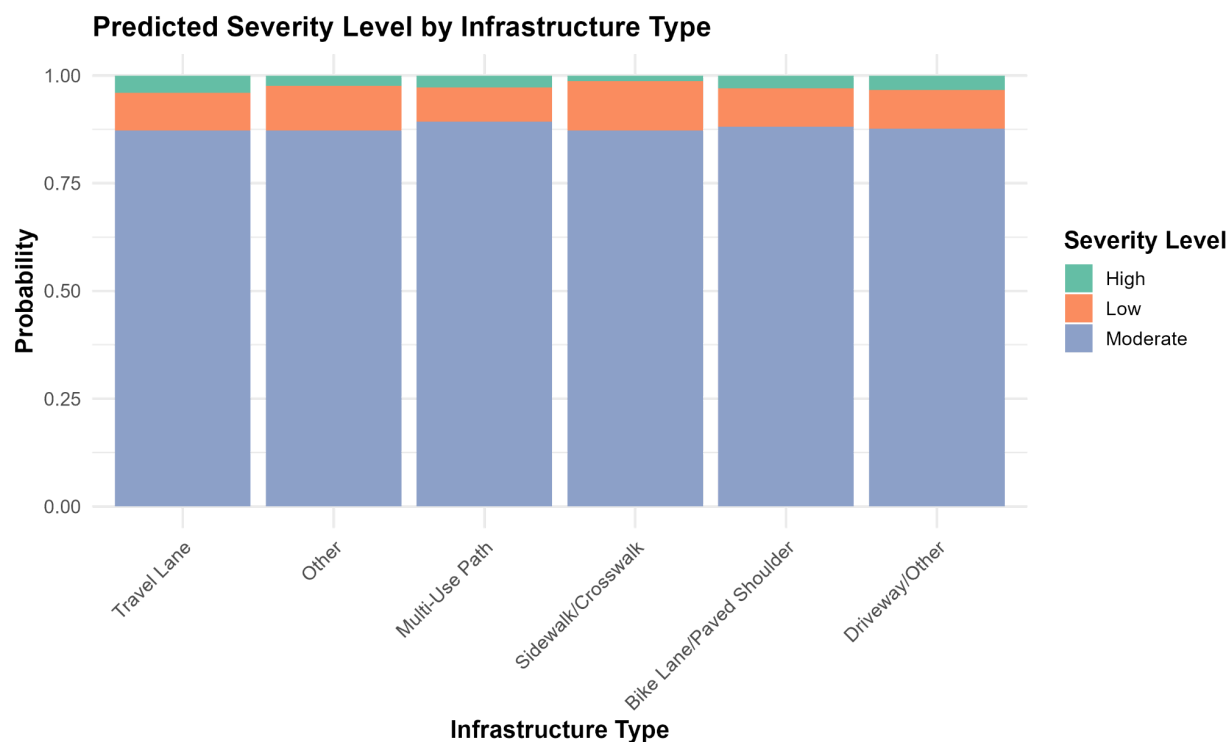
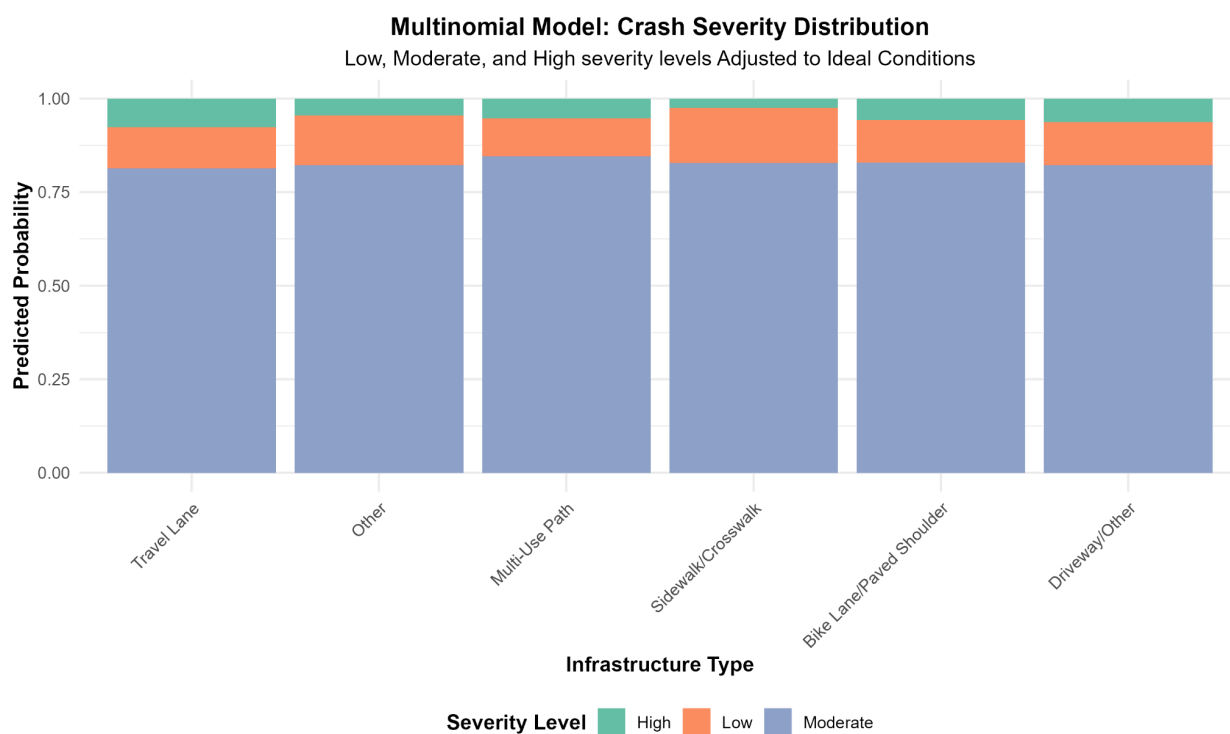


Figure Thirteen - Multinomial Logistic Regression Adjusted to Ideal Conditions (ggpredict)



The multinomial regression model (Figure 11) showed a significant connection between infrastructure type and crash severity outcome. In particular, crashes at sidewalks and crosswalks had the most substantial effect in protecting bicyclists from medium-severity injuries, with a coefficient of $\beta = -0.26$, an OR of 0.77, and a p-value less than 0.001. Travel lanes, which served as the reference category, were linked to the highest severity risk.

Figures 12 and 13 provide insight into how crash severity probabilities shift under ideal conditions. When adjusted for factors such as daylight, clear weather, dry pavement, and local road classification, sidewalks and crosswalks showed the highest proportion of low-severity crashes at 14.8%, compared to 11.1% for travel lanes. Travel lanes continued to show the highest probability of high-severity outcomes, at 7.6%, while sidewalks remained the lowest at 2.5%. Comparing adjusted to unadjusted conditions reveals that low-severity crash probabilities increased by two to three percentage points across most infrastructure types. For example, travel lanes increased from 8.8% to 11.1%. At the same time, moderate-severity outcomes declined, while high-severity outcomes remained largely stable, changing by less than one percentage point. This suggests that ideal environmental conditions primarily shift moderate injuries toward lower severity, rather than significantly reducing the likelihood of the most severe crashes.

d. Binomial Regression Model

Figure Fourteen - Binomial Logistic Regression Summary

	Estimate	Standard Error	z value	Pr(> z)	
(Intercept)	-3.235	0.099	-32.697	0.0000	***
Simplified_Pos2Other	-0.567	0.163	-3.473	0.0005	***
Simplified_Pos2Multi-Use Path	-0.365	0.427	-0.854	0.3933	
Simplified_Pos2Sidewalk/Crosswalk	-1.179	0.134	-8.774	0.0000	***
Simplified_Pos2Bike Lane/Paved Shoulder	-0.298	0.124	-2.399	0.0164	*
Simplified_Pos2Driveway/Other	-0.182	0.167	-1.088	0.2768	
LightCondDark - Lighted Roadway	0.378	0.096	3.943	0.0001	***
LightCondDark - Roadway Not Lighted	0.783	0.082	9.597	0.0000	***
LightCondDusk	0.508	0.153	3.320	0.0009	***
LightCondOther	0.301	0.212	1.418	0.1561	
WeatherCloudy	0.013	0.105	0.126	0.8996	
WeatherFog, Smog, Smoke	0.925	0.434	2.134	0.0329	*
WeatherRain	-0.159	0.219	-0.724	0.4690	
WeatherOther	-0.786	0.774	-1.015	0.3103	
RdClassNc Route	0.958	0.100	9.596	0.0000	***
RdClassSecondary Route	0.685	0.084	8.135	0.0000	***
RdClassUs Route	0.863	0.103	8.373	0.0000	***
RdClassOther	-0.901	0.344	-2.616	0.0089	**
RdSurfaceCoarse Asphalt	0.162	0.063	2.579	0.0099	**
RdSurfaceOther	-0.134	0.233	-0.575	0.5652	
RdConditioUnknown	-0.100	0.769	-0.130	0.8970	
RdConditioWet	-0.002	0.169	-0.011	0.9909	
RdConditioOther	0.205	0.458	0.447	0.6550	
CrashLocIntersection-Related	0.114	0.126	0.908	0.3640	
CrashLocNon-Intersection	0.423	0.070	6.015	0.0000	***
CrashLocOther	-11.872	197.970	-0.060	0.9522	
NumLanesLow (1-2)	0.116	0.070	1.654	0.0981	.

Signif. codes: 0 '***' < 0.001 < '**' < 0.01 < '*' < 0.05

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 8454 on 14158 degrees of freedom

Residual deviance: 7791 on 14132 degrees of freedom

(1414 observations deleted due to missingness)

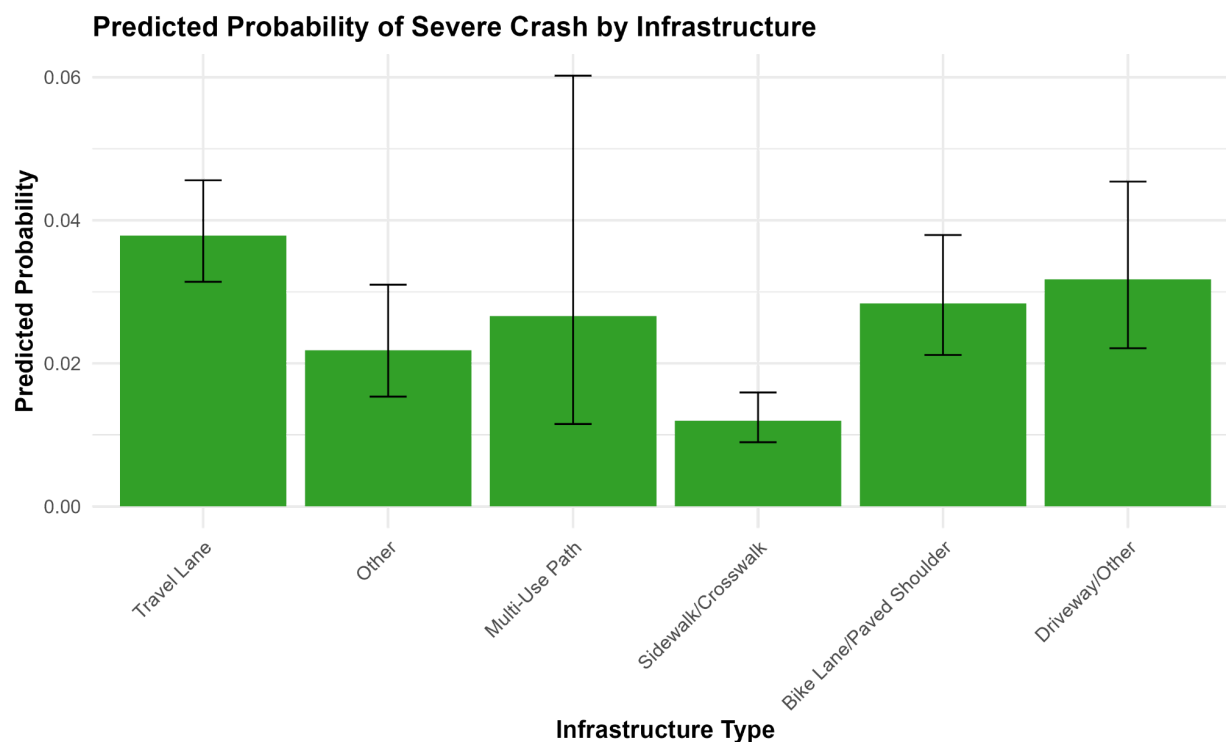
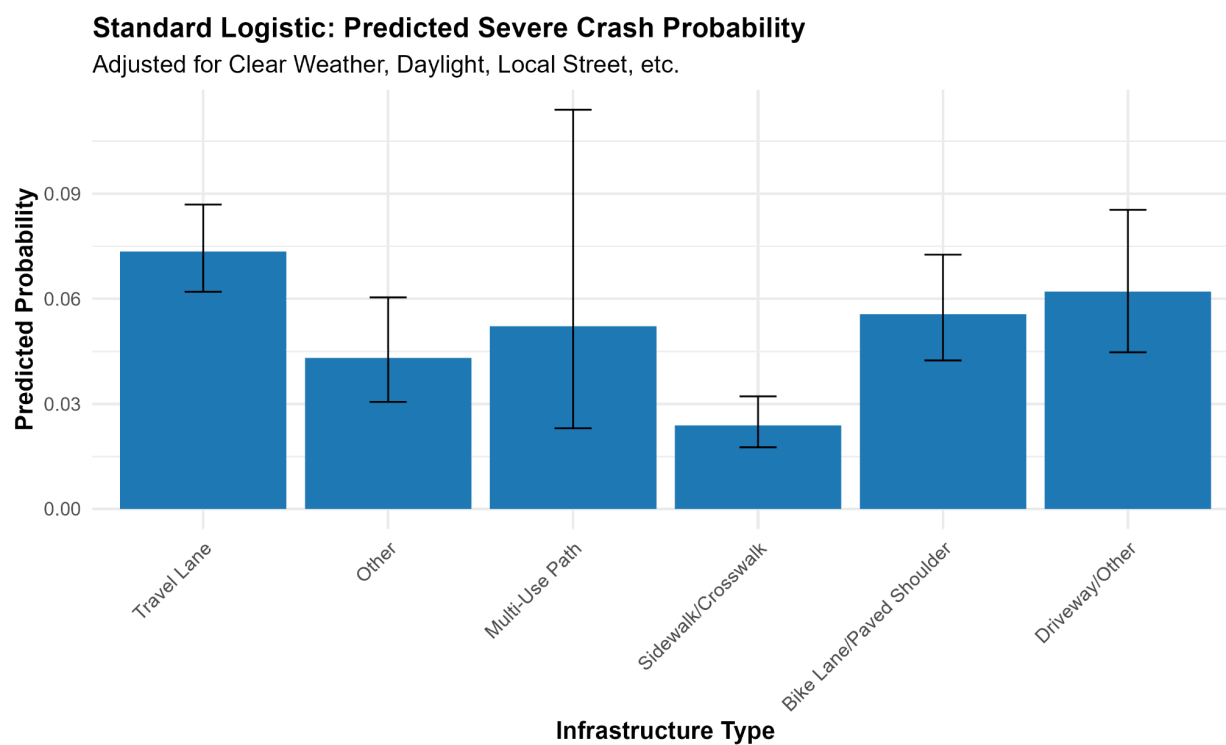
Figure Fifteen - Binary Logistic Model Results (ggpredict)*Figure Fifteen - Binary Logistic Model Adjusted to Ideal Conditions (ggpredict)*

Figure Seventeen - High Severity Percentage: Interaction Between Infrastructure and Location

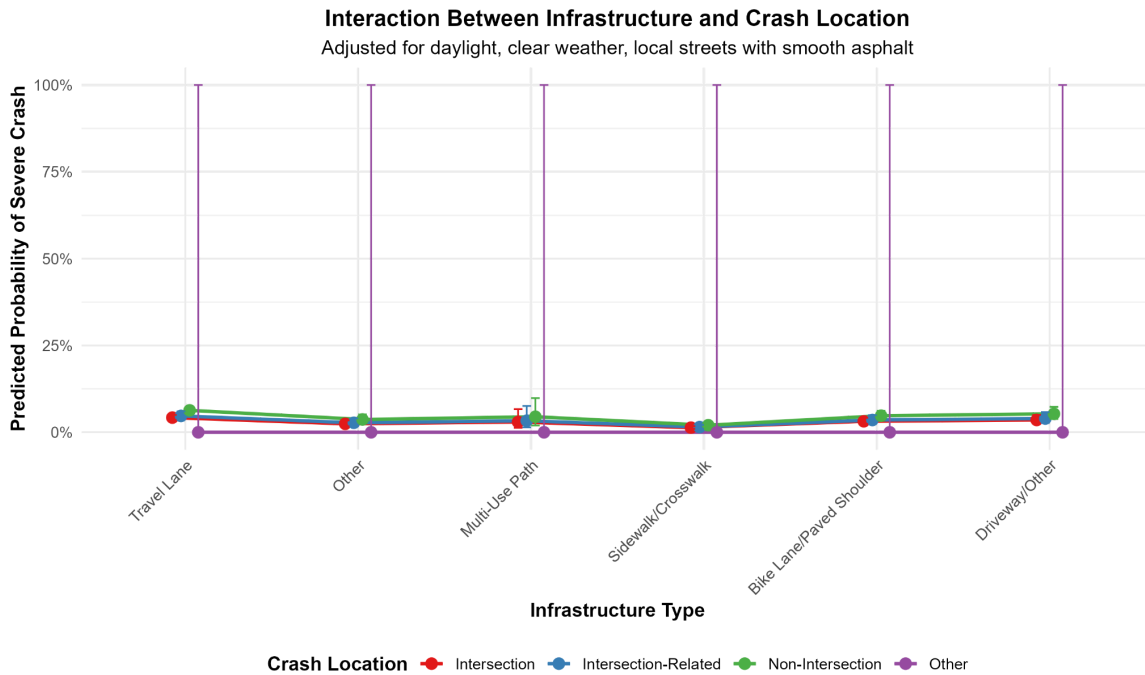


Figure Eighteen - High Severity Probability: Interaction Between Infrastructure and Rd. Condition.

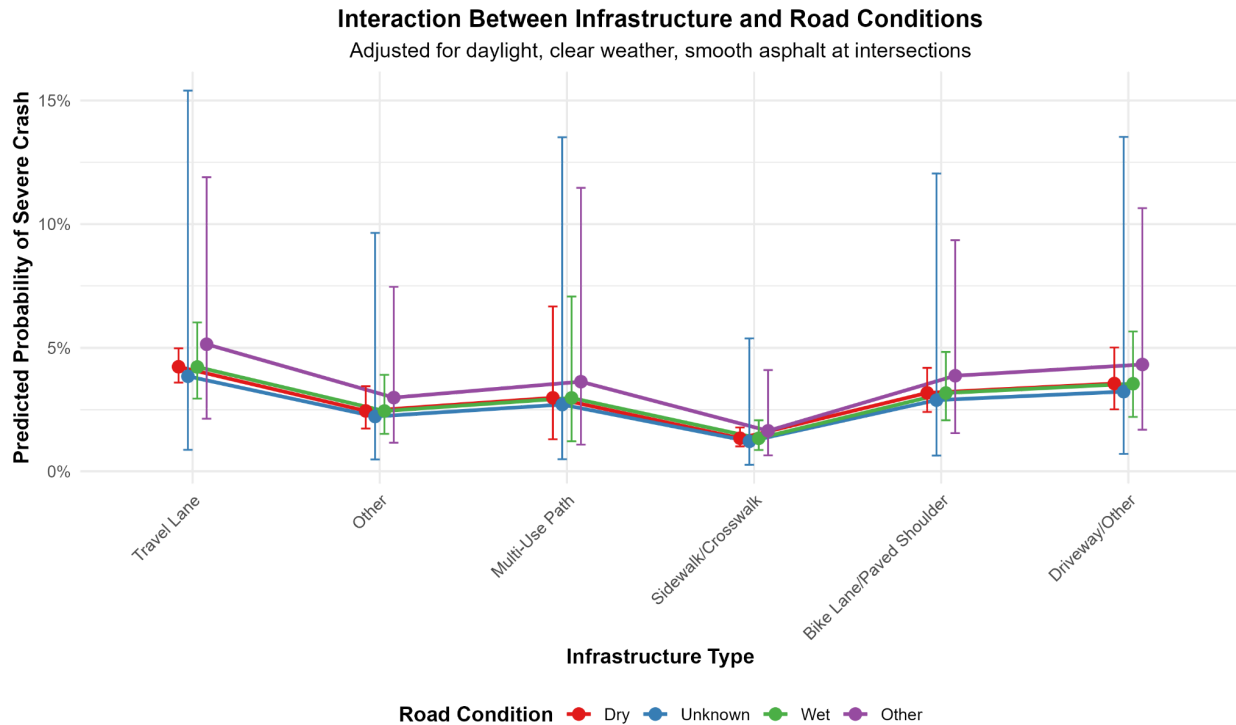


Figure Nineteen - High Severity Percentage: Interaction Between Infrastructure and Rd. Surface

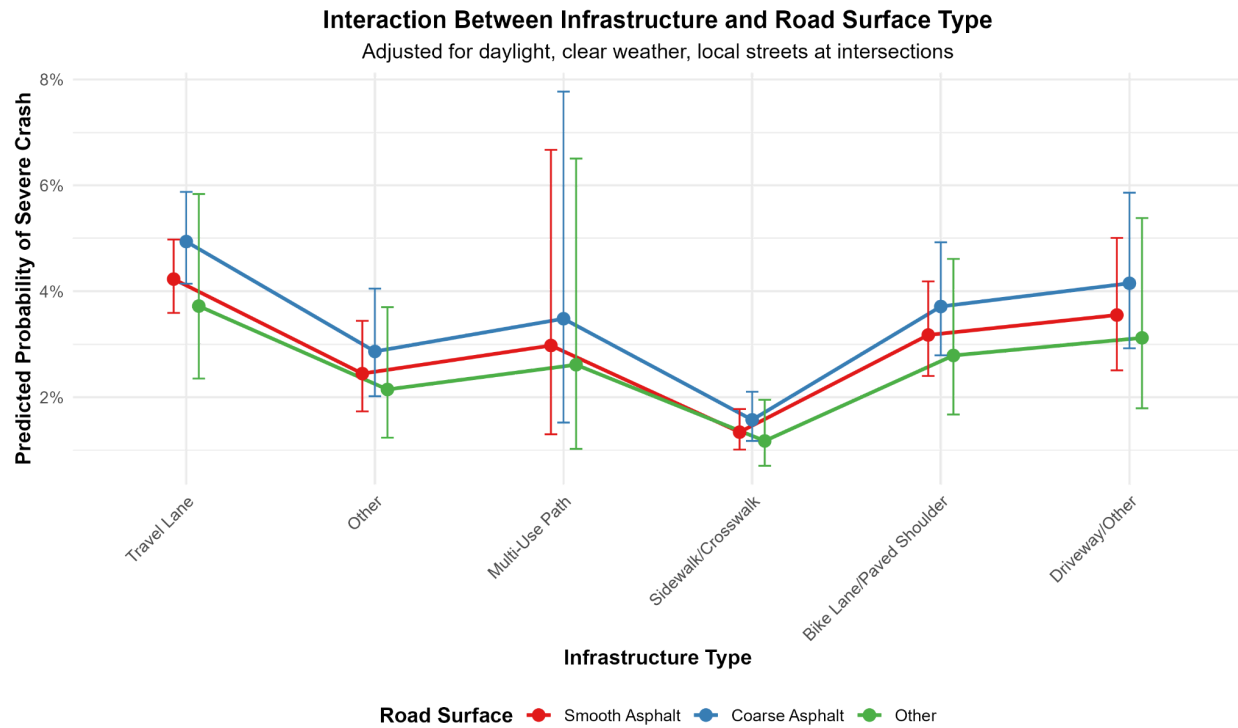


Figure Twenty - High Severity Percentage: Interaction Between Infrastructure and Light

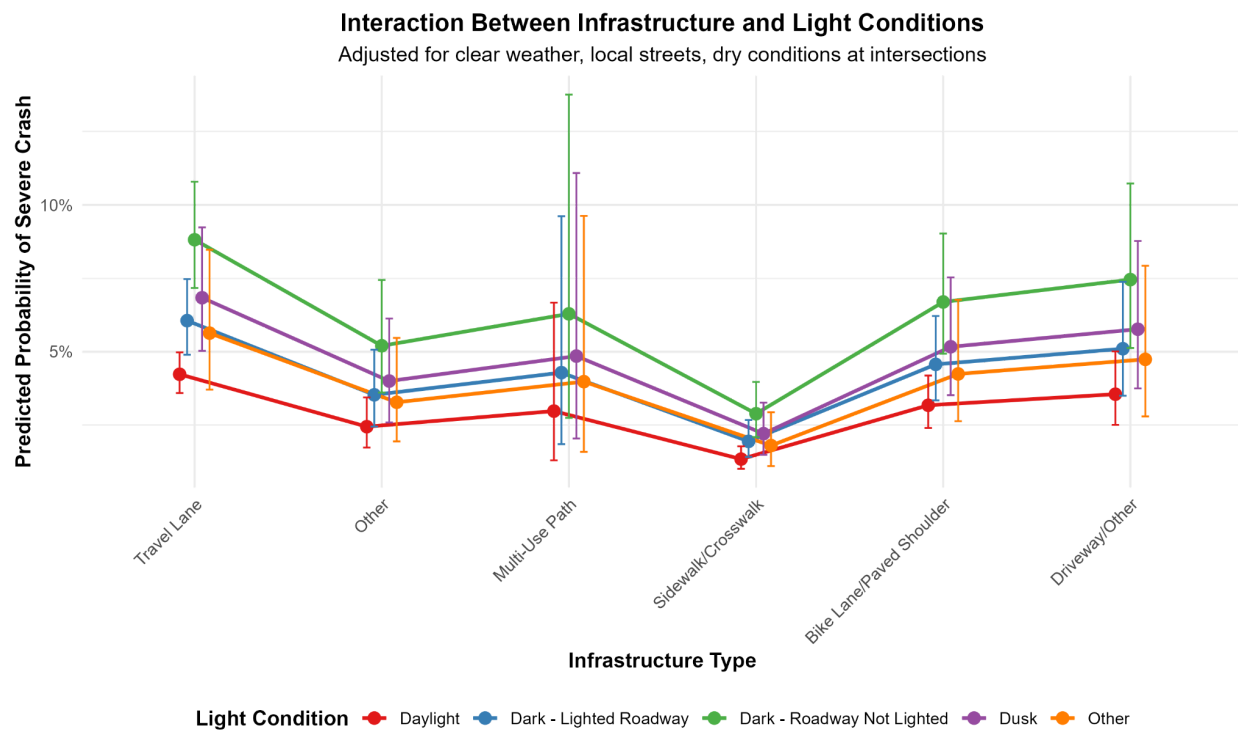


Figure Twenty-One - High Severity Percentage: Interaction Between Infrastructure and Location

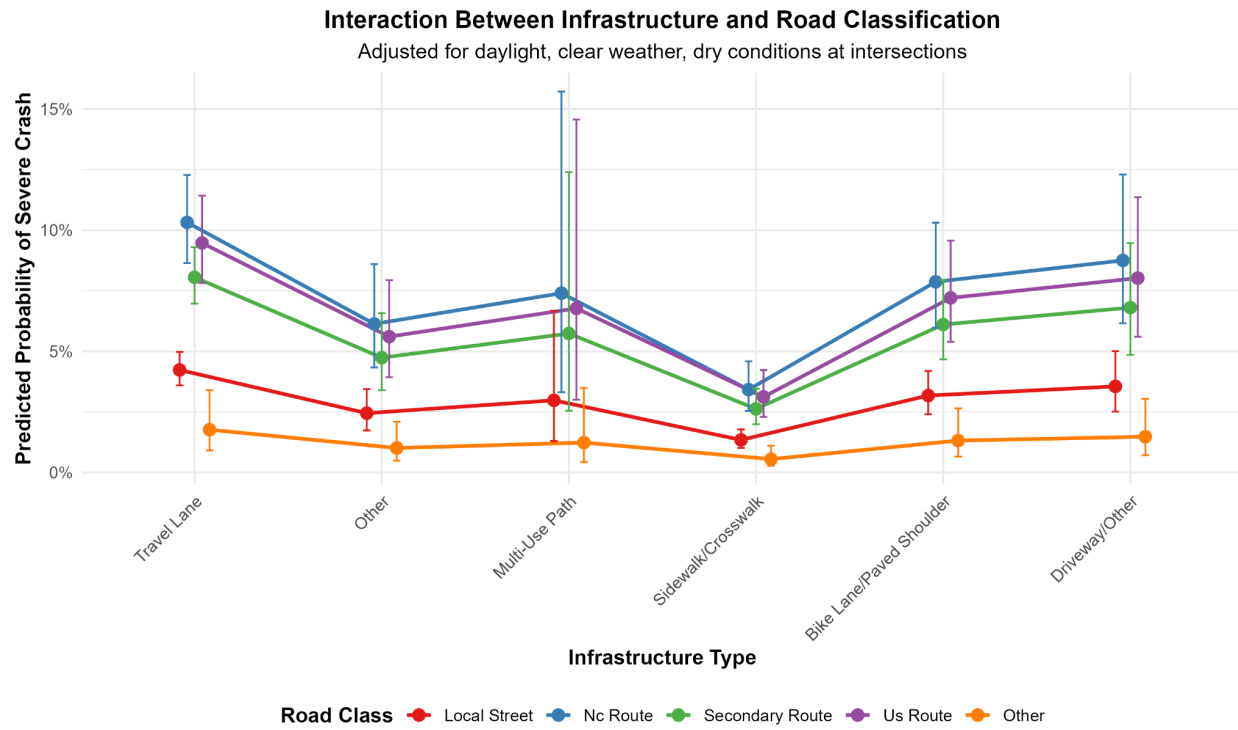


Figure Twenty-Two - High Severity Percentage: Interaction Between Infrastructure and Weather

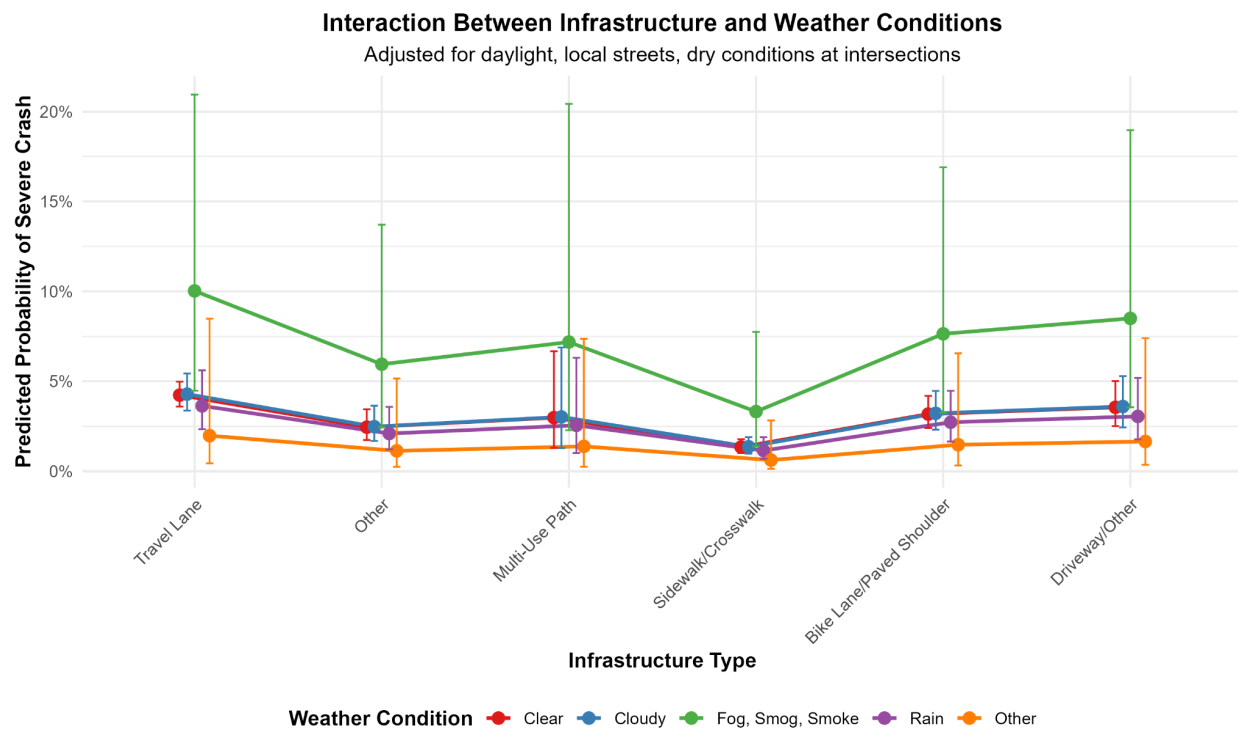


Figure Twenty-Three - Heat map comparison between Infra. And Rd. Class (Severe Only)

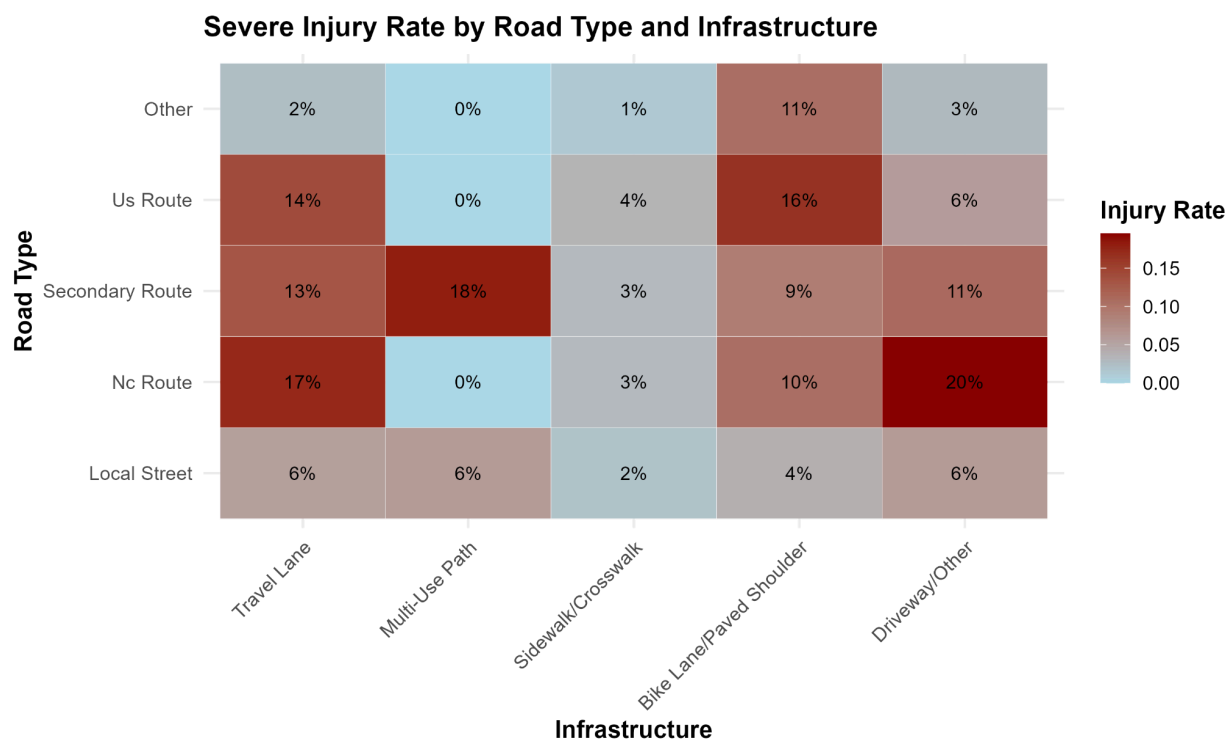
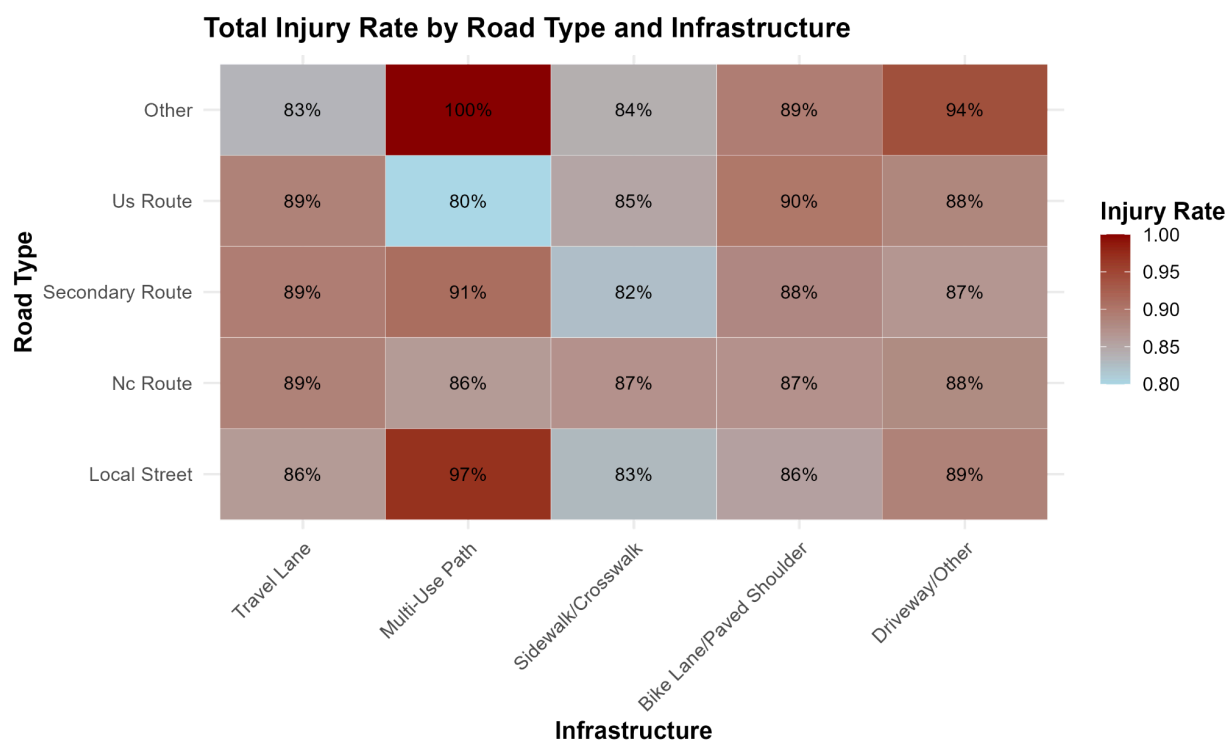


Figure Twenty-Four - Heat map comparison between Infra. And Rd. Class (All Injuries)



The binomial logistic regression analysis (Figure 11) identified multiple external factors that were linked to the likelihood of a crash resulting in severe injury. Sidewalks and crosswalks were the most effective at reducing risk, cutting the chances of a severe injury by about 69% compared to travel lanes (OR = 0.31, 95% CI [0.24, 0.40], $p < 0.001$). Bike lanes and paved shoulders also reduced risk, though to a lesser extent, with a 26% decrease in severe crash odds (OR = 0.74, 95% CI [0.58, 0.94], $p = 0.016$). Crashes that occurred on unlit roadways at night had 2.19 times higher odds of severe injuries from crashes compared to those occurring in daylight (OR = 2.19, 95% CI [1.86, 2.57], $p < 0.001$). Road classification, surprisingly, proved to be one of the strongest factors overall with NC routes had more than 2.6 times the odds of severe crashes relative to local streets (OR = 2.61, 95% CI [2.14, 3.17], $p < 0.001$), followed by US routes (OR = 2.36, 95% CI [1.93, 2.90], $p < 0.001$), and secondary roads (OR = 1.99, 95% CI [1.70, 2.34], $p < 0.001$). Additionally, coarse asphalt was associated with an 18% increase in the odds of a crash causing a severe injury compared to smooth asphalt (OR = 1.18, 95% CI [1.04, 1.33], $p = 0.010$). Interestingly, crashes that occurred away from intersections were more likely to be severe than those at intersections (OR = 1.53, 95% CI [1.33, 1.75], $p < 0.001$). As for weather conditions, only fog, smog, or smoke showed a significant association with severity, with more than double the odds of a severe outcome (OR = 2.53, 95% CI [1.08, 5.92], $p = 0.033$), though the limited observations for this factor caused some instability in the CI. The model explained about 8% (pseudo- $R^2 = 0.079$) of what makes some crashes worse than others, which might sound low, but that's pretty standard for crash data. Every crash is different, and a lot depends on random factors that can't be measured, like how fast someone was going or exactly how the collision happened. Nonetheless, the most influential factors were infrastructure type, road classification, and lighting conditions. Other variables like rainy weather ($p = 0.469$),

wet surfaces ($p = 0.991$), and number of lanes ($p = 0.098$) were not statistically significant after controlling for the other factors in the model.

Figures 15 through 22 show how different variables affect the probability of severe crashes across various types of bicycle infrastructure. Each figure highlights the impact of one specific factor on crash severity, helping to illustrate how conditions like lighting, weather, or road type interact with different facilities. Additionally, Figures 15 and 16 showcase the difference in the probability of severe crashes under raw probability (Figure 15) and under adjusted probability with ideal conditions (Figure 16). The outcome had adjusted probabilities higher across all infrastructure types, but relative risk patterns remained consistent. Travel lanes saw the largest increase in severe crash probability, rising from 3.8% unadjusted to 7.3% adjusted. Sidewalks and crosswalks remained the safest, with only a modest increase from 1.2% to 2.4%.

Figures 23 and 24 further dissect these patterns through heatmap comparisons of severe and total injury rates across infrastructure-road type combinations. Figure 23 (severe injuries only) had NC routes paired with driveway/other to have the highest severe injury rate (20%), while local streets with Sidewalks/Crosswalks had the lowest non-zero rate (2%). Notably, Secondary routes had elevated severe crash risks across most infrastructure types (3%-18%). Figure 24 (all injuries) tells a different story, with injury rates clustering between 80–100% regardless of road class, highlighting how nearly all crashes on certain facilities (multi-use path and driveways/other) result in some injury, even if not severe. Together, these visualizations confirm that while injury likelihood is widespread, severe outcomes are disproportionately concentrated on higher-order roads (US/secondary routes) and specific infrastructure combinations.

E. Discussion

This study's comprehensive analysis of North Carolina's bicycle crashes reveals several key facts that both support and challenge existing literature on bicycle infrastructure safety. The findings indicate that infrastructure type significantly influences crash severity outcomes, but this relationship is heavily moderated by environmental conditions and road classification.

The analysis demonstrates that sidewalks and crosswalks are associated with the lowest probability of high-severity crashes (1.3%) compared to travel lanes (8.8%), which supports Teschke et al.'s (2012) finding that sidewalks can improve safety in areas lacking dedicated bicycle infrastructure. This makes sense, as sidewalk crashes are likely between other pedestrians while travel lane crashes are likely between bicyclists and motor vehicles. Thus, leading to more high-severity crashes. However, this contrasts with Wachtel and Lewiston's conclusion that bicyclists are 1.8 times more likely to be injured on sidewalks compared to roadways. Even when considering crashes as a whole, sidewalks (18.3%) hold $\frac{1}{3}$ of the total crashes of travel lanes (18.3%). This difference could come from the lack of exposure data or the scale of the dataset, as the Wachtel and Lewiston study only observed 41 crashes.³² Additionally, this echoes claims by Hee et al., who reported that self-inflicted falls and low-speed incidents are common in separated infrastructure like sidewalks, thus leading to more crashes at lower severity.

Bike lanes and paved shoulders showed moderate protective effects (9.0% high-severity crashes), offering a 26% reduction in the odds of severe injuries compared to travel lanes. This finding aligns with the growing consensus that designated cycling infrastructure enhances safety, though not as dramatically as some advocates might expect. The relatively high crash rate per mile for bike lanes (0.4 crashes per mile) compared to sidewalks (0.16 crashes per mile) suggests

³² Alan Wachtel and Diana Lewiston, "Risk Factors for Bicycle-Motor Vehicle Collisions at Intersections," *ITE Journal* 64, no. 9 (September 1994): <https://www.bicyclinglife.com/Library/Accident-Study.pdf>. Pp 8.

that while bike lanes reduce severity, they may not necessarily reduce crash frequency. However, this should be taken with a grain of salt, as bike lanes had only 5.9% of the total crash data and are likely exposed to a higher volume of bicyclists due to the attraction of such facilities. Further research on exposure data is necessary to test this claim.

The most surprising, and likely the most flawed, result is the extraordinarily high crash rate for multi-use paths (1.84 crashes per mile), which could potentially support Hee et al.'s argument that increased bicycle volume in dedicated facilities could inflate crash rates. However, the truth of this result is likely that the PBIN's public nature underrepresents the amount of mapped multi-use path locations, as it is highly unlikely that almost every mile constructed would lead to two crashes. This just requires more updated and more accurate data to see if it suffers from both the lack of exposure and underrepresentation.

The binomial regression revealed that road classification is among the strongest predictors of crash severity, with NC routes associated with 2.6 times higher odds of severe crashes compared to local streets. This finding extends beyond most existing literature, which typically focuses on infrastructure type without adequately controlling for road classification. Night-time crashes on unlit roads more than doubled the odds of severe injuries ($OR = 2.19$), while adverse weather conditions like fog increased severity odds by 2.5 times. Finally, and most interestingly, road surface type had a significant impact on crash severity, with coarse asphalt associated with an 18% increase in the odds of severe injury compared to smooth asphalt. This finding suggests that infrastructure designers should consider not only the design of the facility but also the materials used in its construction. If there were more observations on both sand and gravel crashes, it would have been interesting to see their impact as well. These findings build

upon Teschke et al. and Crompton et al, which identified environmental variables primarily as control factors rather than examining their interactive effects with infrastructure.

The spatial analysis revealed interesting patterns, with urban areas like Charlotte and Raleigh showing "cold spots" for severe crashes while rural counties like Robeson and Johnston emerged as hot spots. Meaning that there is a disproportionate amount of high-severity crashes in rural settings compared to their urban counterparts. This could be due to factors such as more available bicycle infrastructure and better lighting in urban areas compared to rural areas. This can be seen through the Kernel density maps (Figures 4-5), which showcase that urban centers like Charlotte and Raleigh contain a majority of both bicycle and pedestrian infrastructure. This spatial distribution challenges common assumptions that urban areas with higher traffic volumes would see a higher proportion of severe crashes. Instead, it suggests that lower-density areas with likely higher speed limits and less infrastructure may pose greater risks for cyclists. Thus, local policymakers need to create designated infrastructure for bicyclists, just like Lincoln County, that will help reduce the severity of the crashes.

Just to reiterate, multiple restraints limited the full scope of this study. First, the reliance on police-reported crashes likely underrepresents minor incidents and those occurring without motor vehicle involvement, potentially biasing severity estimates. Second, the PBIN data quality issues limited the depth of infrastructure analysis possible. From potentially over-representing sidewalks to underrepresenting multi-use paths, the PBIN dataset would need better quality checks, especially since it is crowd-sourced. Third, the absence of reliable exposure data prevents calculation of true risk rates, as crash frequencies cannot be normalized by ridership levels across different infrastructure types. Future research should prioritize gaining access to private datasets such as the North Carolina Non-Motorized Volume Data Program or personally

conducting a state-wide exposure count. However, this might be hard to do as bicycle exposure data is one of the main challenges faced in bicycle safety analysis, not only for this study but also for studies like Alluri et al.³³ Additionally, incorporating more variables like cyclist experience, helmet use, and cycling speed would create a more sound model with more of the variance in the crash severity being explained. These variables would be quite easy to gain, as multiple variables within the crash dataset were left unused due to time restraints. Using a negative binomial model would also be useful, as you could get physical counts of crashes per segment or area and model the frequency of severe crashes while accounting for overdispersion in the data.

F. Conclusion

In conclusion, this analysis advances our understanding of bicycle safety beyond the traditional focus on infrastructure type alone. By incorporating environmental factors, road classification, and spatial patterns, our study demonstrates that while dedicated bicycle infrastructure generally enhances cyclist safety, its effectiveness is strongly influenced by external conditions. Thus, suggesting that comprehensive safety improvements for bicyclists require attention to both "the place" and "the pieces" that shape cycling experiences across North Carolina. However, further research must be conducted with both more accurate data and available exposure data to address current gaps in understanding true crash risks and to support the development of more robust and effective risk models.

³³Priyanka Alluri et al., *Statewide Analysis of Bicycle Crashes* (Tallahassee, FL: Florida Department of Transportation, 2017), Pp. 142.

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